[INCH-POUND]
A-A-2936
November 27, 1995
SUPERSEDING
GGG-R-180
January 21, 1992

COMMERCIAL ITEM DESCRIPTION

REAMER, HAND, BURRING, PIPE

The General Services Administration has authorized the use of this Commercial Item Description for procurement of the Type VII, Classes 1 and 3 reamers specified in GGG-R-180 for all federal agencies.

1. SCOPE

This Commercial Item Description covers pipe burring hand reamers used in removing burrs from the inside edges of pipes, for countersinking, and for enlarging holes in sheet metals.

2. CLASSIFICATION

Class 1 - Ratchet type.
Class 2 - Bit brace shank.

3. SALIENT CHARACTERISTICS

3.1 Class 1.

3.1.1 Design. The Class 1 hand reamer shall consist essentially of a lever, handle, ratchet mechanism, and a reamer. Except for the lever, the complete unit shall be made of carbon steel. The class I reamer shall be similar to figure 1 and shall conform to table 1.

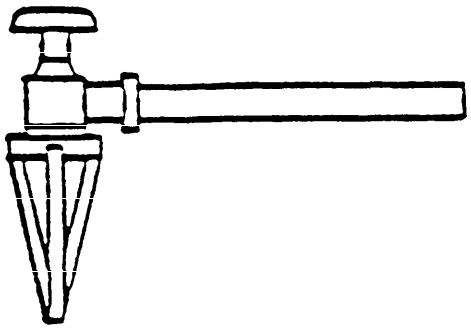


Figure 1. Class 1, pipe burring (ratchet type) hand reamer

Beneficial comments, recommendations, additions, clarification, etc. and any data which may improve this document should be sent to: general Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5110

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3.1.1.1 Lever. The lever shall be made of cast iron or steel, shall be solid or hollow, and shall afford a comfortable grip for the operator. The lever shall be made integral with the ratchet collar or shall be provided with a means for being securely fastened to the ratchet collar.

3.1.1.2 Ratchet mechanism. The ratchet mechanism shall consist of a body with ratchet teeth, a pawl of steel hardened and held in contact with the teeth, and a ratchet collar. The teeth shall be enclosed within the ratchet collar. The reamer shall operate in a clockwise direction.

3.1.1.3 Handle. A suitable handle shall be provided, directly in line with the reamer, to enable the operator to support and guide the reamer during operation. The handle shall rotate about the ratchet body and reamer, and shall be attached in a manner to preclude its working loose while in use.

3.1.1.4 Reamer. The reamer shall consist either of a body fitted with three or more hardened and ground blades, or of a solid cone body with either straight or spiral flutes, as specified (see 7.2). The reamer shall be hardened and tempered to a Rockwell hardness value from C54 to C58. When required to be tested, the reamer shall be tested for hardness in accordance with ASTM E18.

Pipe capacity	Distance from the centerline of reamer to end of lever	Weight (complete unit including reamer)
Inches	(minimum) Inches	(maximum) Pounds
1/8 to 2	10	7 3/4
3/8 to 3	10	9 1/2

Table 1. Class 1, pipe burring (ratchet type) hand reamer

3.2 Class 2.

3.2.1 Design. The Class 2 reamer shall have a bit brace shank and consist of a ground and solid cone body with either straight or spiral flutes, as specified (see 7.2). The Class 2 reamer shall be made of carbon steel and shall be hardened and tempered to a Rockwell hardness value from C54 to C57. When required to be tested, the reamer's hardness shall be tested in accordance with ASTM E18. The Class 2 reamer shall taper 20 to 30 degrees (included angle), and shall have a minimum of 8 to a maximum of 12 flutes. The Class 2 reamer shall be similar to figure 2 and shall conform to table 2.

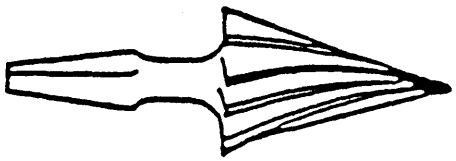


Figure 2. Class 2, pipe burring (bit brace shank) hand reamer

Pipe capacity (Inches)	Diameter (inches)	Diameter (Inches)
	At point	At large end
1/8 to 1	3/16	1 1/4
1/4 to 1 1/4	1/4	1 15/32
1/4 to 2	1/4	2 17/64

Table 2. Class 2, pipe burring (bit brace shank) hand reamer

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3.3 Interchangeability. All parts shall be designed and manufactured to definite standards, clearances, and tolerances in order that any such parts of particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit.

3.4 Performance. The reamers shall be capable of hand reaming two samples each of pipe within the capacity of the size of reamer specified. The finish of the countersink in the sample pipes shall be smooth and free of burrs.

3.5 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g. sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety or appearance.

3.6 Identification marking. The reamer shall be marked with the manufacture's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the Contracting Officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall provide certification and maintain substantiating evidence that all items being offered for acceptance meet the requirements of this Commercial Item Description and conform to the manufacturer's drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

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5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING

Requirements for preservation, packing and marking shall be as specified in the contract or order.

7. NOTES

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of reference documents.

Federal Specifications: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC, 20407.

Federal Acquisition Regulations (FAR): Government Printing Office Superintendent of Documents, Washington, DC 20401-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 43201-4606.

ASTM E18. American Society for Testing and Materials, 1916 race Sty., Philadelphia, PA 19103.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and shall include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) Class and pipe capacity.
(c) When applicable, the appropriate NSN.
(d) Whether straight or spiral flutes are required.
(e) If bid samples are required and the number of samples required.
(f) If first article samples are required and the number of samples required.
(h) If lot sampling inspections are required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Examinations. Each sample tool shall be examined for all salient characteristics. The inspection Level shall be S-3 with an Acceptable quality level (AQL) of 2.5 percent defective.

Testing. Each sample tool shall be tested for all salient characteristics. The inspection Level for the hardness test shall be S-2 with an Acceptable quality level (AQL) of 2.5 percent defective. The inspection Level for the performance test shall be S-2 with an Acceptable quality level (AQL) of 4.0 percent defective.

(i) What preservation, packaging, and marking are required.

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National Stock Numbers (NSNs). NSNs shown may not be indicative of all possible NSNs associated with this CID:

NSN	Class	Flutes
5110-00-186-4215	1	Straight
5110-00-222-9040	1	Spiral

Military Interests:
e: DoD has no registered interest in revisions and
ices to thisCommercial Item Description until further notice.

Preparing Activity:
GSA - FSS

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